

A cost and emission optimization framework for strategic intermodal freight transportation infrastructure development

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ABSTRACT

The freight transportation sector is critical to the economic prosperity in the US, but it also constitutes a major source of carbon emissions. Although intermodal transportation has been shown to increase operating efficiency and reduce carbon emissions, research on infrastructural support for intermodal transportation is still insufficient. In this study, we establish a mixed integer programming model to jointly optimize strategic infrastructure development decisions and freight transportation decisions over a long horizon. Our model features a mixture of traditional single-mode facilities and hybrid hubs that facilitate rail–water transportation integration. A branch-and-cut decomposition algorithm is developed to solve large-scale problems. We collect real-world freight, infrastructure, and operations data to conduct computational studies on the model performance and algorithm efficiency. We provide insights for practitioners to address infrastructure planning and budgetary concerns. A case study using the established model at the national scale shows that well-optimized transportation infrastructure investment could have over 300% return during a 25-year horizon. In addition, fully capitalizing on the maturing clean vehicle technologies could reduce carbon emissions by 73.56 million tons at an annual rate.

1. Introduction

The transportation sector is a cornerstone of the US economy. According to the Bureau of Transportation Statistics, it contributes \$1.7 trillion, or 6.9%, to the total U.S. GDP in 2022 [1]. However, it is also one of the major contributors to U.S. Greenhouse Gas (GHG) emissions. In 2022, it is responsible for 28% of total emissions in the U.S., according to the Environmental Protection Agency (EPA) [2]. Within the transportation sector, freight transportation generates 8%–11% of total GHG emissions and is expected to nearly triple by 2050 [3].

Freight transportation systems typically include five main modes: road, rail, water, air, and pipeline [4]. Road transportation, which primarily consists of trucking, dominates the U.S. freight industry, with about 72.6% of freight transported by trucks in 2022 [5]. However, trucking is the second most significant GHG emitter among other modes. According to the Congressional Budget Office, the emissions from shipping one ton of cargo for one mile are approximately 0.17 kg for trucks, compared with 0.10 kg for trains and 0.04 kg for ships [6]. Besides the environmental impacts, railway and waterway transportation are more cost-effective than highway. Transportation costs associated with the railway are about 25% of highway costs; waterway costs are about 5% of highway costs [7].

This makes intermodal transportation a promising approach to enhance transportation cost-effectiveness and reduce emissions. Intermodal networks integrate highways, railways, and waterways to introduce efficiency and flexibility, reducing transportation costs for bulk cargo over long distances [8]. Intermodality has also been shown to be a key strategic element in reducing GHG emissions [9]. Integrating intermodal networks with freight consolidation strategies significantly reduces environmental impacts [10].

The term multimodal transportation is also widely used in the literature. While multimodal transportation is sometimes used interchangeably with intermodal transportation, they represent different approaches in the context of logistic operations. In an intermodal system, freight is moved in a single loading unit across multiple modes without handling the cargo during transfers, ensuring container integrity throughout the journey [11]. In contrast, multimodal transportation refers more broadly to the use of multiple modes for the same shipment, but the cargo may be reloaded, handled, or repacked at various stages [12]. In this study, we focus on intermodal systems exclusively due to their practical relevance. Many intermodal facilities are already in place thanks to their long-haul efficiency, enabling seamless mode shifts while reducing cargo damage and processing time [13,14].

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Although much effort has been made to study intermodal transportation, significant gaps still exist in the literature. As pointed out, proper infrastructure and terminal design are critical to a successful intermodal system, which ensures maximum economic and environmental benefits [11]. However, the joint infrastructure development and freight transportation optimization remain under-investigated. Further, existing research mainly focuses on small-scale case studies, pilot projects, or theoretical frameworks, with only a few large-scale real-world implementations [15]. This study addresses the methodological gaps in the literature in the following ways. First, many existing models are limited to regional applications on small scales and depend on off-the-shelf optimization solvers. In contrast, our framework is designed for national-level analyses and incorporates a tailored Benders decomposition algorithm with an integrated cutting plane procedure to enhance computational tractability and improve scalability. Second, in the literature, few studies have explicitly accounted for the multi-year interdependency in infrastructure planning. By modeling infrastructure development over a multi-year horizon, we capture the timing, location, and sequencing of infrastructure investments in a way that better reflects real-world decision-making processes. Third, most prior approaches separate strategic infrastructure development decisions from freight flow decisions. Our method integrates infrastructure development and freight flow decisions within a unified optimization model to globally optimize heterogeneous decisions. Last but not least, hybrid intermodal facilities have yet to be widely considered in the literature for large-scale strategic planning. Our model presents a comprehensive approach to studying the real-world impact of integrated intermodal infrastructure.

Specifically, we introduce a strategic intermodal infrastructure development framework over a long planning horizon. We model nationwide intermodal freight transportation in the continental U.S. We optimize over existing highway, railway, and water infrastructure, at the same time considering constructing hybrid hubs that facilitate freight flow across three modes. Hybrid hubs have been shown to improve transportation efficiency, decreasing reliance on trucking for long-distance hauls, and supporting emission reduction efforts [16,17]. Our main contributions are as follows:

- We develop a Mixed Integer Programming (MIP) model to jointly optimize intermodal infrastructure development and freight transportation decisions over a long planning horizon.
- We design an efficient decomposition solution algorithm by exploiting the MIP model structure and applying state-of-the-art optimization theory.
- We conduct comprehensive experiments to validate model versatility under varying parameters and compare algorithm performances under varying problem sizes.
- We curate a highly detailed U.S. nationwide intermodal dataset, including network infrastructure, freight volume, capacity, cost, and emission data, to project intermodal freight transportation and carbon emissions during a 26-year horizon. The projection provides insights to practitioners regarding infrastructure development, hybrid hub effectiveness, and clean vehicle technology investment.

In the following, Section 2 reviews the literature on intermodal transportation networks and hub location models. Section 3 outlines the methodology, including the optimization model and the decomposition algorithm. Section 4 describes the data collected and utilized to support the findings in this study. Section 5 conducts computational studies, covering sensitivity analyses, algorithmic comparison, and real-world large-scale case studies. Section 6 concludes this study and offers recommendations for future research. Appendix A further summarizes all abbreviations in this study.

2. Literature review

Intermodal transportation has been studied by many over the past decades [18]. The majority of the research focuses on the transportation of containers between road, rail, and water networks. Air freight, although prevalent in the transportation industry, is typically excluded from the intermodal network, due to its high carbon emissions and expensive costs [19,20]. In contrast, studies have shown that a well-integrated road, rail, and water transportation network can effectively promote cost-efficiency and reduce carbon emissions [21,22]. As such, in this study, we focus on the freight transportation among road, rail, and water networks.

Many existing studies optimize intermodal transportation from an operational perspective [23,24]. Typically, a regional network is established with a limited amount of cargo [25]. Vehicles, such as trucks, trains, and ships, are modeled in detail [23]. This greatly benefits practical adoptions, as practitioners can identify the routes and schedules of intermodal services. However, operational-level modeling significantly limits the scale of the problem before it becomes intractable. Computational complexity of the model often grows exponentially with the size of the network, making nationwide applications difficult [26]. Operational studies mostly optimize over a short time horizon, e.g., weekly or monthly transportation demand [8,27]. Simulation has been a widely adopted approach to study intermodal transportation systems [28]. For example, Zhao et al. [29] utilized to model routing and congestion under real-time conditions. This brings a challenge when considering transportation infrastructure development, as infrastructure decisions have a prolonged impact over an extended time horizon.

As such, when investigating system-wide behaviors, a strategic perspective becomes necessary. In the literature, strategic intermodal facility location plays a critical role. Well-sited facilities promote mode switching, consolidation, and sorting functions, which are vital to intermodal transportation efficiency [12,30,31]. Studies have also highlighted that infrastructure planning improves both economic and environmental performance, aligning with long-term stakeholder goals [32]. These strategic decisions are often made jointly with operational ones. Several studies show that separating strategic and operational layers can lead to poor outcomes. For example, Heggen et al. [33] found that treating long-haul and local routing separately reduced overall efficiency. Elbert et al. [30] demonstrated that combining hub location with service network design yields more realistic solutions and reduces costs by over 3%. Lin and Lin [34] employed a two-stage model to distinguish between strategic and operational choices while still allowing them to interact. Alumur et al. [19] also emphasized the need to align strategic planning with time-sensitive service constraints. In the same direction, Yıldız et al. [12] proposed a model that integrates hub decisions with routing and capacity planning. Bakker et al. [35] investigated a decarbonization and fuel uncertainty problem through a stochastic model, focusing on policy-level scenarios and assumes that terminals are already in place.

One of the promising strategic intermodal infrastructure improvement approaches is the construction of hybrid hubs. Hybrid hubs enable the integration of multiple transportation modes within a single station, offering many operational and strategic advantages in intermodal network design. In the literature, Dai et al. [31] demonstrates that hybrid configurations, combining road, rail, and inland waterway links allow for more efficient routing and lower environmental impacts by supporting flexible network structures beyond traditional hub-and-spoke systems. Similarly, Li et al. [36] highlights how hybrid hubs in rail and road freight systems improve service quality and infrastructure utilization, particularly when supported by hierarchical hub structures. Furthermore, Wang and Meng [37] demonstrates that hybrid terminals enhance container routing efficiency by accommodating diverse operator preferences and routing behaviors, particularly in discrete intermodal planning frameworks. In addition, Ziaei and Jabbarzadeh [17] emphasizes that hybrid hubs increase network resilience

Table 1

A summary of intermodal transportation studies that consider infrastructure development decisions.

Study	Modes	Cost metrics	Scale & region	Hybrid hubs	Method	Decision level
Dai et al. [31]	Road, rail, water	Operation, infrastructure	Small, China	Yes	MIP	Strategic, operational
Derpich et al. [24]	Road, rail, water	Operation, emission, infrastructure	Medium, Chile	No	MIP	Strategic
Elbert et al. [30]	Road, rail	Operation, infrastructure	Medium, Germany	No	MIP	Strategic, operational
Fattahi et al. [44]	Road, rail	Operation, risk, infrastructure	Medium, Iran	No	Heuristics	Strategic, operational
Gallardo et al. [45]	Road, rail	Emission, infrastructure	Small, New Zealand	No	Simulation	Strategic
Li et al. [36]	Road, rail	Operation, infrastructure	Medium, China	Yes	Heuristics	Strategic, operational
de Morais et al. [46]	Road, rail, water	Operation, emission, infrastructure	Medium, Brazil	No	MIP	Strategic
Teye et al. [47]	Road, rail, water	Operation, infrastructure	Small, Australia	No	MIP	Strategic
Wang and Meng [37]	Road, rail, water	Lateness, infrastructure	Medium, –	Yes	MIP	Strategic
Yildiz et al. [12]	Road, rail	Operation, infrastructure	Medium, Turkey	No	Decomposition	Strategic, operational
Ziaei and Jabbarzadeh [17]	Road, rail, water	Operation, emission, lateness, infrastructure	Medium, –	Yes	Robust optimization	Strategic
Zhao et al. [48]	Rail, water	Operation, infrastructure	Small, China	No	Heuristics	Strategic
This study	Road, rail, water	Operation, emission, infrastructure	Large, US	Yes	Decomposition	Strategic

by enabling mode switching under uncertainty, which is particularly beneficial for robust planning in multimodal logistics. Collectively, these studies underscore the strategic value of hybrid hubs in enhancing flexibility, reducing emissions, and improving service performance in complex intermodal systems.

To efficiently make strategic and operational decisions on system-wide intermodal transportation, network flow models are widely adopted [7,38]. Intermodal facilities are represented by nodes in networks, and connections, such as highway, railway, and waterway, are represented by arcs in networks. Cargo transportation is then modeled as flows through the network. Freight flows can be either continuous or discrete, depending on cargo types. Continuous flows represent weight or tonnage, often used for bulk commodities, and allow continuous decision variables [39,40]. Discrete flows are better suited to containerized freight, where cargo must be handled in indivisible units, such as Twenty-foot Equivalent Units (TEUs), typically requiring mixed-integer formulations [41,42]. Some models, like [41], combine both approaches using receding horizon optimization to capture bulk and container trade-offs. Network modeling also enables robustness assessment to study how the system reacts to disruptions, providing insights of system resilience [43].

Externalities, especially GHG emissions, have been increasingly considered alongside cost metrics when evaluating intermodal efficiency. As mentioned, the transportation sector accounts for a significant amount of GHG emissions [1], and intermodal transportation has been shown to be effective in reducing emissions [8,9]. To address this, both single- and multi-objective approaches have been widely used for intermodal transportation. Multi-objective models minimize transportation costs and GHG emissions in separate objective functions [11]. These models highlight the tension between economic and environmental goals by generating Pareto fronts. Multi-objective formulations are also adopted to capture risks, especially in sensitive applications like hazardous material logistics [17]. However, multi-objective approaches make it difficult to evaluate cost and emission metrics in a uniform way, due to different scales and units. In contrast, single-objective models

mostly use carbon taxes to calculate the monetary value of emissions, making it intuitive in evaluating the trade-offs between operating costs and emissions [7]. This also reduces the complexity as single-objective models do not require the additional computation of Pareto fronts.

As real-world transportation networks grow more complex, commercial solvers that implement generic algorithms become insufficient in solving large-scale problems. Decomposition methods, such as Benders decomposition, have been increasingly used to derive efficient algorithms to solve large-scale mixed-integer programs. Decomposition methods have addressed computational issues in many transportation research areas, such as port operations [49], drone-assisted delivery [50], vehicle routing [51], scheduling [52], etc. In intermodal transportation, decomposition has been utilized as a major solution approach, but it is mostly adopted to solve operational-level problems [23,53], whereas intermodal infrastructure has seen limited usage [12]. This has partially undermined the capability in conducting nationwide studies of intermodal infrastructure development. Benders decomposition is well-suited for solving intermodal transportation problems, since it effectively handles heterogeneous decisions in a hierarchical structure [54]. Particularly, in intermodal infrastructure problems, Benders decomposition can fully separate the facility location problem from the network flow problem, adopting a divide-and-conquer strategy to efficiently solve large-scale instances over long horizons. This study will integrate Benders decomposition into generic commercial solvers to improve computational efficiency and enable large-scale case studies.

Table 1 provides a comparative overview of recent studies. The table focuses specifically on studies that incorporate infrastructure development decisions that address strategic-level decision-making. It summarizes key characteristics, including the modes of transportation, types of cost components, e.g., operating, GHG emissions, infrastructure, geographic and spatial scales, use of hybrid hubs, solution methods, and planning levels. The diversity in modeling approaches and cost structures reflects a growing emphasis on environmental sustainability. Notably, only a few studies consider hybrid hubs integration or

Table 2
Definition of parameters and variables.

Parameters	
Name	Description
$M = \{H, R, W\}$	Set of modes, where H, R, W denote highway, railway, and waterway.
$G^m(V^m, A^m)$	Network of mode m , where $m \in M$.
$V^F = V^P \cup V^R \cup V^W$	Set of facilities, including hybrid, rail, and water nodes.
$V^C \subseteq V^F$	Set of candidate locations to construct rail, water, or hybrid facilities.
T	Set of times, e.g., $T = \{1, 2, 3, \dots\}$.
$(\gamma_o, \gamma_d, \gamma_w) \in \Gamma_t$	Cargo to be transported in year t with origin γ_o , destination γ_d , and weight γ_w , where Γ_t is the set of cargo.
C^m	Transportation costs for mode $m \in M$ (\$/ton.mile).
c_i	Operations costs for hub $i \in V^F$ (\$/ton).
δ^m	GHG emission costs of transporting using mode $m \in M$ (\$/ton.mile).
σ_i	GHG emission costs for hub $i \in V^F$ (\$/ton).
$d_{i,j}^m$	Distance between hubs i and j where $m \in M$ (miles).
B_t	Available budget for period $t \in T$ (\$).
$b_{t,i}$	Construction cost for hub $i \in V^m$ where $m \in \{R, W\}$ at $t \in T$ (\$)
$f_{t,i}$	Fixed cost for hub $i \in V^m$ where $m \in \{R, W\}$ at $t \in T$ (\$)
g_i	Capacity of hub $i \in V^m$ where $m \in \{R, W\}$ (ton).
$k_{i,j}^m$	Capacity of arc $(i, j) \in A^m$ where $m \in \{R, W\}$ (ton).
Variables	
Name	Description
$X_{t,i,j,o}^m \in \mathbb{R}_+$	Amount of freight to be transported on arc (i, j) from origin $o \in V^H$ at time $t \in T$ by mode $m \in M$ (ton).
$Y_{t,i} \in \{0, 1\}$	If hub i is built at time $t \in T$.
$Z_{t,i} \in \{0, 1\}$	If hub i is available at time $t \in T \setminus \{1\}$.

combine multiple decision layers within a strategic planning horizon. Most existing studies consider only small- to medium-scale networks, whereas we conduct large-scale case studies at the national level over a long horizon. This underscores the novelty and relevance of our work, which bridges these gaps through a flexible, cost- and emission-integrated strategic framework that optimizes intermodal infrastructure development decisions.

3. Method

In this section, we first develop an MIP model to optimize intermodal infrastructure development and cargo transportation jointly over a long horizon. We then apply Benders decomposition to derive an algorithm that solves the problem efficiently.

3.1. MIP

The model consists of four sets of nodes. Warehouse nodes V^H are accessible by road transportation through highways and serve as the origins and destinations of cargo. Rail hubs V^R are accessible by road and rail transportation. Water ports V^W are accessible by road and water transportation. Potential hybrid hubs V^P are accessible by all modes (road, rail, and water). We define highway, railway, and waterway networks as directed graphs $G^H(V, A^H)$, $G^R(V^R, A^R)$, and $G^W(V^W, A^W)$. All nodes are included in the highway network, where $V = V^H \cup V^R \cup V^W \cup V^P$. The railway network connects rail hubs and potential hybrid hubs, where $V^R = V^R \cup V^P$. The waterway network does the same, containing all the nodes in V^W , where $V^W = V^W \cup V^P$. We further let $V^F = V^P \cup V^R \cup V^W$ be the combined facility set of hybrid, rail, and water nodes. We define single-direction arcs for the networks in sets A^H , A^R , and A^W , representing physical connections between intermodal facilities, including highways, railways, and waterways, respectively.

The model identifies the optimal set of hubs to be constructed and operated over a specific period. This represents strategic decisions that will impact operations during a time horizon T , where $t \in T = \{1, 2, \dots\}$ can be considered as one year in practical applications. Facility location decisions are made over a set $V^C \subseteq V^F$, which represents candidate nodes where a rail, water, or hybrid facility can be built. Let $Y_{t,i} \in \{0, 1\}$, $t \in T, i \in V^C$ indicates the decision to build a facility at location i at the beginning of the period t . Let $Z_{t,i} \in \{0, 1\}$, $t \in T, i \in V^F$ be the availability of hub i at time t , i.e., $Z_{t,i} = 1$ if

there is a facility available at location i at the beginning of the time period t . We assume that if $Y_{t,i} = 1$, the facility is available at the beginning of the period $t + 1$, i.e., $Z_{t,i} = 0$ and $Z_{t+1,i} = 1$. In addition, we set $Z_{1,i} = 0$, $i \in V^C$, so that no candidate facility is available at the beginning of the horizon. In contrast, we set $Z_{1,i} = 0$, $i \in V^F \setminus V^C$. As such, this study primarily considers improvement decisions at the nodes of the intermodal network. Edge improvements are not included since the U.S. highway and freight railroads infrastructure is relatively complete and transportation capacity bottlenecks are often identified at hubs [55,56]. Waterway edges further depend on natural geographical features, e.g., rivers and coastal regions, making intermodal facilities the primary target for infrastructure development.

The infrastructure development decisions impact freight transportation during the horizon T . We define $\gamma = (\gamma_o, \gamma_d, \gamma_w) \in \Gamma_t$ as the cargo to be transported in year t , with origin γ_o , destination γ_d , and weight γ_w in tons, where Γ_t represents the set of cargo in year t . The freight flow in each arc $(i, j) \in A^m$, originated from the origin $o \in V^H$, during period t by mode $m \in M$ is modeled by a continuous decision variable $X_{t,i,j,o}^m \in \mathbb{R}_+$, which represents the tons of cargo transported. The index o is added to ensure the transportation through $X_{t,i,j,o}^m$ is consistency with the demand γ at origins and destinations. The choice of modeling continuous flows reflects the nature of publicly available data and the focus on national-scale freight movements aggregated over long periods. Since the flows represent the total weight between hubs, rather than individual container movements, continuous modeling is a more appropriate and efficient approach than integer variables. It also reduced the computational complexity compared with integer formulations, allowing the model to focus on strategic infrastructure development decisions and system-wide cost and emission optimization.

Three types of costs are included in the model: operational costs, GHG emission costs, and infrastructure development costs. Operational costs include cargo transportation costs and hub operating costs. For transportation cost, we define C^m (\$/ton.mile) as the cost of transporting one ton of cargo for a mile by mode $m \in M$. For hub operating cost, we define c_i (\$/ton) as the cost at hub $i \in V^F$ for handling one ton of cargo. We further include a fixed cost for hub f_i , $i \in V^C$, which represents costs such as rent and utility, incurred once hub construction is completed. The cost for building a hub at location $i \in V^C$ is b_i (\$). We define B_t (\$), $t \in T$ as the budget for building hubs in the period t . Similar to operational costs, GHG emission costs consist of transportation and hub emissions. We use δ^m (\$/ton.mile)

to denote the emission cost associated with transporting one ton of cargo for a mile by mode $m \in M$. We use σ_i (\$/ton) to denote the emission at hub $i \in V^F$ for handling one ton of cargo. Note that when estimating the parameters, emission costs are calculated by the emitted CO₂ equivalent times the carbon tax. In addition to costs, each hub in the network, whether existing or candidate, has a specific capacity denoted by g_i (tons). This capacity represents the cargo tonnage that can be handled by hub i in a specific period $t \in T$. The capacity at each edge in the railway and waterway networks is defined as $k_{i,j}^m$, $\forall (i,j) \in A^m$, $m \in \{R, W\}$. This represents the maximum amount of cargo that each railway and waterway segment can handle annually. We do not restrict the roadway network under the assumption that all transportation demand can be met using the highway network, although this may lead to great inefficiencies in the transportation sector. The above notation is summarized in Table 2. As such, the MIP model is formulated as follows:

$$\min \sum_{t \in T, i \in V^F} \left[f_{t,i} Z_{t,i} + \sum_{\substack{m \in M, o \in V^H \\ (i,j) \in A^m}} (c_i + \sigma_i) X_{t,i,j,o}^m \right] \\ + \sum_{\substack{t \in T, m \in M \\ o \in V^H, (i,j) \in A^m}} (C^m + \delta^m) d_{i,j}^m X_{t,i,j,o}^m \quad (1a)$$

$$\text{s.t.} \sum_{i \in V^F} b_{t,i} Y_{t,i} \leq B_t \quad \forall t \in T \quad (1b)$$

$$Y_{t,i} \leq 1 - Z_{t,i} \quad \forall i \in V^F, t \in T \quad (1c)$$

$$Z_{t-1,i} + Y_{t,i} \leq Z_{t,i} \quad \forall i \in V^F, t \in T \setminus \{1\} \quad (1d)$$

$$\sum_{\substack{m \in M, o \in V^H \\ (i,j) \in A^m}} X_{t,i,j,o}^m \leq g_j Z_{t,j} \quad \forall t \in T, j \in V^F \quad (1e)$$

$$\sum_{m \in M, o \in V^H} X_{t,i,j,o}^m \leq k_{i,j}^m \quad \forall t \in T, (i,j) \in A^m, m \in \{R, W\} \quad (1f)$$

$$\sum_{j \in V} X_{t,i,j,i}^H = \sum_{\gamma \in \Gamma_i, \gamma_o = i} \gamma_w \quad \forall t \in T, i \in V^H \quad (1g)$$

$$X_{t,i,j,i}^H + \sum_{h \in V^F} X_{t,h,j,i}^H = \gamma_w \quad \forall t \in T, i, j \in V^H, i \neq j, \gamma \in \Gamma_i, \gamma_o = i, \gamma_d = j \quad (1h)$$

$$\sum_{\substack{j \in V \\ (i,j) \in A^H}} X_{t,i,j,o}^H + \sum_{\substack{j \in V^R \cup V^P \\ (i,j) \in A^R}} X_{t,i,j,o}^R = \sum_{\substack{j \in V \\ (i,j) \in A^H}} X_{t,i,j,o}^H + \sum_{\substack{j \in V^R \cup V^P \\ (i,j) \in A^R}} X_{t,i,j,o}^R \\ \forall t \in T, i \in V^R, o \in V^H \quad (1i)$$

$$\sum_{\substack{j \in V \\ (i,j) \in A^H}} X_{t,i,j,o}^H + \sum_{\substack{j \in V^W \cup V^P \\ (i,j) \in A^W}} X_{t,i,j,o}^W = \sum_{\substack{j \in V \\ (i,j) \in A^H}} X_{t,i,j,o}^H + \sum_{\substack{j \in V^W \cup V^P \\ (i,j) \in A^W}} X_{t,i,j,o}^W \\ \forall t \in T, i \in V^W, o \in V^H \quad (1j)$$

$$\sum_{m \in M, j \in V, (i,j) \in A^m} X_{t,i,j,o}^m = \sum_{m \in M, j \in V, (i,j) \in A^m} X_{t,i,j,o}^m \quad \forall t \in T, i \in V^P, o \in V^H \quad (1k)$$

$$X_{t,i,j,o}^m \in \mathbb{R}_+ \quad \forall (i,j) \in A^m, o \in V^H, t \in T, m \in M \quad (1l)$$

$$Y_{t,i}, Z_{t,i} \in \{0, 1\} \quad \forall i \in V^F, t \in T. \quad (1m)$$

In the MIP model, the objective function includes fixed facility cost, facility operating costs, facility emission cost, transportation cost, and transportation emission cost. The first part summarizes all costs related to facilities, and the second part summarizes all costs related to transportation activities. Constraint (1b) represents the budget, ensuring that all investment costs for a specific period $t \in T$ are within the allocated budget B_t . Constraints (1c)–(1d) ensure the logical sequencing of hub construction and operating decisions through variables $Z_{t,i}$ and $Y_{t,i}$. Specifically, hubs can only be constructed at candidate locations; only existing hubs can be operated. Once operated, a hub remains operational until the end of the horizon. Constraint (1e) ensures that the total amount of cargo handled by a hub does not exceed its capacity. Constraint (1f) ensures that the cargo transported on each arc does not exceed the capacity of that arc. Constraints (1g)–(1h) ensure that the inflow and outflow at each origin and destination warehouse meet the demand and supply requirements, respectively. The origin index o

ensues that the amount of cargo transported to a destination matches the amount initially dispatched at the origin. Constraints (1i)–(1k) represent the network flow constraints that maintain the balance at hubs, while keeping track of origin-specific flows to ensure consistency. Specifically, for rail stations and water ports, the transportation modes exclude water and rail; for hybrid hubs, all modes of transportation are available.

As such, in this study, we develop the model primarily considering governmental institutions as the decision-maker, since the public sector has been a major contributor in transportation infrastructure development projects [57,58]. The developed method will assist decision-makers in selecting the optimal infrastructure locations, allocating funding, estimating sector-wide impact, and reducing carbon emissions through legislation. These capabilities are shown throughout the following sections. Nonetheless, the private sector has also been investing in intermodal infrastructure projects [59]. The methods developed in the study can assist individual freight companies in investment efforts as well, to identify bottlenecks in current practices, plan for strategic expansions, and strengthen business operations under emission regulations.

3.2. Benders decomposition

Given the complexity of the MIP due to the binary decision variables $Z_{t,i}$ and $Y_{t,i}$, the problem becomes computationally expensive for large networks. Benders decomposition is an effective solution method to address this [60]. Benders decomposition has been widely adopted by many studies [54]. The central idea is to decompose a large-scale MIP into a Master Problem (MP) that optimizes integer variables and Subproblems (SP) that optimize continuous flow variables.

For the MIP model (1a)–(1m), we decompose the infrastructure development decisions and the cargo flow decisions. The MP can be formulated as

$$\text{MP} = \min_{Y, Z, \Psi} \left\{ \sum_{t \in T, i \in V^F} f_{t,i} Z_{t,i} + \sum_{t \in T} Q_t(Z) : (1b)–(1d) \right\}, \quad (2)$$

where $Q_t(\cdot)$ represent the SP at time t . For each $t \in T$, let $\bar{Z}_{j,t}$, $j \in V^F$, $t \in T$, be feasible solutions from the MP. The SP is then defined as

$$Q_t(\bar{Z}) = \min_X \left\{ \sum_{\substack{i \in V^F, o \in V^H \\ m \in M, (i,j) \in A^m}} (c_i + \sigma_i) X_{t,i,j,o}^m + \sum_{\substack{m \in M \\ o \in V^H, (i,j) \in A^m}} (C^m + \delta^m) d_{i,j}^m X_{t,i,j,o}^m : \right. \\ \left. \sum_{\substack{m \in M, o \in V^H \\ (i,j) \in A^m}} X_{t,i,j,o}^m \leq g_j \bar{Z}_{j,t} \quad \forall j \in V^F, (1f)–(1k) \right\} \quad (3)$$

Essentially, Benders decomposition takes advantage of the two-level decision structure in the MIP: strategic infrastructure development and operational freight transportation. The MP determines the operational status of the existing and constructed hubs across the planning horizon, while the SPs, one per time period $t \in T$, calculate the minimum cost cargo flows given the hub configuration $\bar{Z}$. The objective function of the MP includes the SP values $Q_t(\bar{Z})$, which is approximated by successively adding Benders optimality cuts derived from dual solutions of the SP.

It is important to note that the SPs in this model is always feasible no matter the MP solution. This is because of the structure of the flow conservation constraints. Specifically, constraints (1g)–(1h) ensure that for any MP solution $\bar{Z}$, demand and supply requirements at each warehouse can always be satisfied by the highway network. As such, the SP, $Q_t(\bar{Z})$, $\forall t \in T$, has complete recourse, i.e., the decomposed model does not need feasibility cuts to regular MP solutions.

The optimality cuts are formulated by introducing variables $\theta_t \in \mathbb{R}$, $t \in T$, in the master problem to replace $Q_t(\bar{Z})$. Let $\mu_{t,i}$ be the dual variable corresponding to the hub capacity constraint in the SP (3). For each subproblem at $t \in T$, if $\theta_t < Q_t(\bar{Z})$, the following optimality cuts are added to the MP:

$$\theta_t \geq Q_t(\bar{Z}) + \sum_{i \in V^F} \mu_{t,i} g_i \cdot (Z_{t,i} - \bar{Z}_{i,t}) \quad (4)$$

3.3. A branch-and-cut (B&C) algorithm

To improve the computational efficiency of solving large-scale problems in practice, we integrate Benders optimality cuts into a Branch-and-Bound (B&B) algorithm for integer problems. This combination is particularly effective for models that involve both strategic and operational decisions, as it leverages problem structure to generate valid cuts and reduce solution space. The effectiveness of such a method has been demonstrated throughout the literature in many optimization problems, including transportation networks [54,61].

Algorithm 1: Benders decomposition implemented as a B&C algorithm.

```

1 Initialize the MP with  $\theta_i \geq -N, \forall i \in T, N \gg 0$ ;
2 Set the initial B&B node as  $k = 0$  and the nodal problem as  $MP_k$ ;
3 repeat
4   Solve the linear relaxation of  $MP_k$  and obtain a solution  $(\bar{Z})$ ;
5   if  $(\bar{Z})$  is integer then
6     for  $t \in T$  do
7       Define and solve the SP  $Q_t(\bar{Z})$ ;
8       if  $\theta_t < Q_t(\bar{Z})$  then
9         Obtain the dual multiplier  $\mu_{t,i}$  from  $Q_t(\bar{Z})$ ;
10        Add a cut (4);
11      else
12        Prune node  $k$  or update the bound;
13        Select the next node  $k', k \leftarrow k'$ ;
14      end
15    end
16  else
17    Prune  $k$  or branch from  $k$  according to the B&B
18    tree-searching strategy;
19    Select the next node  $k', k \leftarrow k'$ ;
20  end
21 until all nodes are explored;
```

Specifically, for each node in the B&B tree, if we obtain an integer solution of the variables $Z_{t,i}$ from a linearly relaxed nodal problem, the corresponding subproblem is solved to generate a valid Benders optimality cut (4). An SP is constructed for each time period $t \in T$ based on the obtained integer solution $\bar{Z}_{t,i}$. Each SP determines the minimum flow cost given the operational status of the intermodal hubs. The SPs return the values $Q_t(\bar{Z})$, which is then compared with the solved $\bar{\theta}_t$ from the MP. If the convergence condition $\theta_t \geq Q_t(\bar{Z})$ is not satisfied, an optimality cut (4) is added to improve MP approximations of the SP. Algorithm 1 shows the flow of the B&C algorithm.

Unlike traditional Benders decomposition, where the MP is solved repeatedly at every iteration, Algorithm 1 only solves the MP once. This accelerates the convergence of Benders decomposition while maintaining the solution quality, as shown in the following computational studies. Algorithm 1 also fully takes advantage of state-of-the-art solvers. Additional strategies such as cutting planes and heuristic searches can be readily implemented by the solver to further enhance the solution efficiency.

4. Data

In this study, we collect transportation network, freight volume, and cost data from the continental U.S. Since there is no readily-available data that can be directly utilized by our model, we construct the required data from scratch based on different sources. In the following, we collect and integrate data from road, rail, and water transportation in a central dataset.

4.1. Intermodal networks

The multi-layered intermodal network is constructed by integrating three networks of road, rail, and water transportation. Each network consists of nodes and edges. Nodes represent freight-handling facilities,

including warehouses, rail terminals, and ports. Edges are the connections between two nodes within the same network, including highways, railroads, and inland waterways.

Highway origin and destination nodes are collected from the Freight Analysis Framework (FAF) by the U.S. Department of Transportation [62]. Freight origins and destinations are cleaned and converted to U.S. cities and states. In total, 113 nodes are extracted. The Railway network consists of 56 nodes representing intermodal rail terminals operated by Norfolk Southern and Union Pacific. These nodes are identified using GIS coordinates from public rail maps [13,14]. The waterway network comprises 48 nodes, representing both inland and coastal ports. These locations are selected based on significant domestic waterborne tonnage from the U.S. Army Corps of Engineers and the FAF data [63]. In total, 217 nodes across three modes are collected. The 217-node dataset is further expanded to a 265-node dataset, where water port locations are considered as candidates for hybrid hubs.

The edges between highway, railway, and waterway nodes are constructed using the Freight and Fuel Transportation Optimization Tool (FTOT) [64]. Fig. 1 shows the collected highway, railway, and waterway networks used in this study. Note that we assume that the highway network not only connects origins and destinations, it also connects rail and water facilities. Considering that the entire 256-node network may be computationally challenging, sub-networks consisting of 26, 39, and 80 nodes are further constructed for computational studies. When constructing each sub-network, facilities with the highest annual throughput are prioritized in the selection using publicly available data [13,14,62,63]. The sub-networks are also manually adjusted to geographically cover the region of the continental U.S. The selected sub-network nodes are shown in Appendix B. Connections between nodes are consistent with those plotted in Fig. 1. For each sub-network, the hybrid hub candidates are selected at the same location as the water hubs to ensure waterway access. That is, facilities can be constructed at existing water ports to strengthen their operations with intermodal capabilities.

4.2. Demand

The freight volume data are sourced from FAF, which provides projected freight volumes between major U.S. cities for selected years: 2025, 2030, 2035, 2040, 2045, and 2050. The FAF dataset categorizes volumes by transportation mode, but in this study, the volumes across all modes are aggregated for each origin–destination pair to reflect overall freight demand. This aggregation is consistent with the strategic focus of the model, which aims to optimize infrastructure decisions at the nationwide level rather than mode-specific operations.

Since FAF data are of 5-year intervals, freight volumes in between the years cannot be obtained directly. To construct annual freight volumes, we extrapolate FAF data by fitting linear regression models for each origin–destination pair. Linear regression is utilized due to its simplicity, transparency, and proven effectiveness in capturing long-term trend behaviors in freight demand modeling, especially when detailed short-term dynamics are not the primary focus. Recent studies have demonstrated the applicability of linear regression in freight demand forecasting [65,66].

The assumption that freight volumes approximately follow linear growth trends is reasonable at the strategic planning level, where the emphasis is on infrastructure development decisions over decades, rather than capturing year-to-year fluctuations. Although linear regression may not reflect sudden disruptions or nonlinear effects, it provides a stable and interpretable method for projecting future freight flows when historical and forecast data exhibit smooth incremental trends. Specifically, Fig. 2 illustrates the global freight demand evolution from 2025 to 2050. Data between 5-year intervals in FAF projections are interpolated using linear regression models. Mode-specific volumes are estimated according to FAF. The total volume across all modes

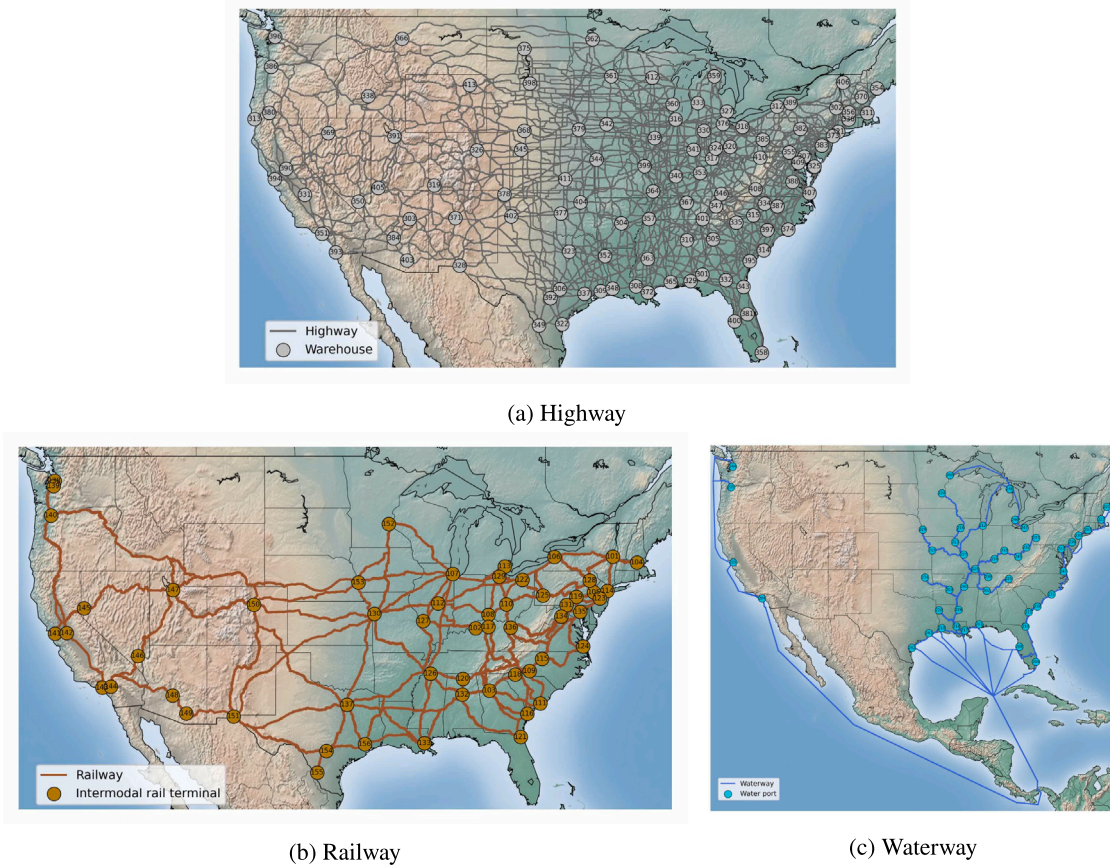


Fig. 1. The 217-node intermodal network.

are used as the demand for each origin–destination pair in following experiments.

From the figures, we can also observe that although the flow patterns over time follow a consistent linear trend across different warehouse pairs, the absolute flow magnitudes vary significantly depending on the freight demand characteristics between different cities. This highlights the spatial heterogeneity in freight volumes across the network, which must be carefully considered in infrastructure planning decisions.

4.3. Model parameters

Transportation costs vary by mode due to differences in infrastructure requirements, fuel efficiency, and operating characteristics. According to the Bureau of Transportation Statistics in the U.S. Department of Transportation, the average freight cost per ton-mile in the US is approximately \$0.15 for trucking [67], \$0.04 for rail [68], and \$0.01 for inland waterways [69]. These values reflect the cost advantage of rail and water transportation over trucking, especially for long-distance, high-volume freight.

The edge capacity of the railway network is estimated using the car-load waybills from the Federal Railroad Administration, which shows that the majority of the U.S. railroads handles up to 100 million tons of freight annually [70]. The edge capacity of the waterway network is difficult to estimate due to the lack of data. Considering that inland container ships have more than three times the capacity of freight trains, and that river transportation is not restricted by infrastructure such as rail lines, we use 300 million tons as the annual capacity [70,71].

Hub construction and operation costs are estimated based on findings from previous academic studies [72,73]. Note that the construction

costs of intermodal facilities are difficult to estimate, which depend on infrastructure size, site conditions, regional construction cost factors, etc. Here, we utilized limited data available in the literature and a real-world project to set the construction cost to be \$1–\$5 billion per facility [73–75]. Fixed annual operating costs, which include facility maintenance, labor, energy consumption, and administrative expenses, generally fall within the range of \$0.5 million to \$1.5 million per year for active terminals. These estimates align with observed costs in U.S. freight infrastructure development studies [72,73]. It is important to note that cost parameters are not static over time. Inflation has been identified as a crucial factor in transport infrastructure projects [76]. An annual inflation of 3% is further incorporated into all costs in the objective function, including fixed facility cost, facility operating costs, facility emission cost, transportation cost, and transportation emission cost. The number is calculated from the averaged 10-year U.S. inflation rates from 2015 to 2024 [77].

To generate consistent and realistic GHG emissions, we utilized the Greenhouse Gases, Regulated Emissions, and Energy Use in Transportation Model (GREET) software [78]. It provides emissions factors and energy consumption estimates for different transportation modes and facility operations. We choose GREET over other emission models in the literature because GREET offers readily available open-source data, computes emissions for all three modes of transportation (i.e., trucks, trains, and ships), and adopts life-cycle analyses to produce well-to-wheel emissions [79,80]. Specifically, heavy-duty vehicles emit 1641 grams of CO₂ equivalent per mile, locomotives emit 26 grams of CO₂ equivalent per ton-mile, and container ships emit 16 grams of CO₂ equivalent per ton-mile. All units are consistent with those used in the GREET data. We estimate carbon tax as \$0.00005 per gram to allow the translation of environmental impacts into monetary cost values [81].

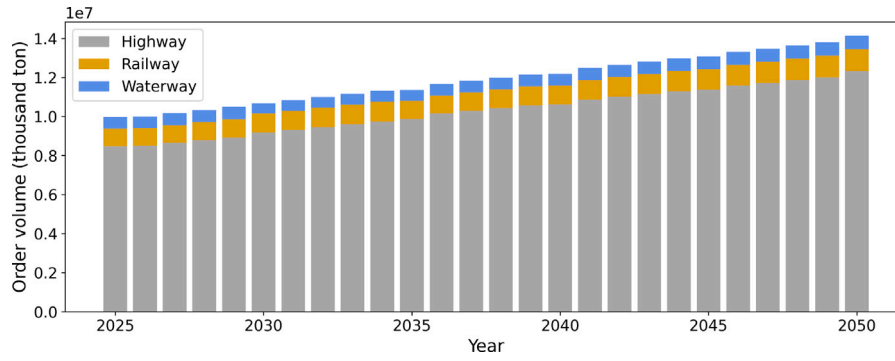


Fig. 2. Global freight demand evolution from 2025 to 2050 estimated from the FAF data [62]. Data between 5-year intervals are interpolated using linear regression models. Mode-specific volumes are estimated according to FAF. The total volume across all modes are used as the demand in following experiments.

Intermodal facility emissions are estimated from the literature and the carbon tax to be \$0.00385 per ton of freight [82]. We further incorporate the International Energy Agency's (IEA) Net Zero by 2050 Analysis to estimate the maturity of alternative clean energy sources [83].

5. Computational studies

In this section, we conduct numerical experiments to validate model and algorithm performances. First, we evaluate algorithm efficiency using different network scales. Then, we establish baseline and investigate model sensitivity under varying parameters. We further run benchmarking experiments to show the value of mode shifts and hybrid hubs. Finally, we highlight the positive impact of clean energy technology using projected data. All experiments are conducted on an Apple iMac with M3 CPU and 64 GB RAM. The Gurobi solver version 12.0.0 is used to solve the models through the Python 3 interface.

5.1. Algorithm comparison

To evaluate the computational performance of the B&C algorithm, we solved instances generated from 26-node, 39-node, 80-node, and 256-node datasets to compare with MIP as the benchmark. For 26, 39, and 80-node networks, five instances are generated to mitigate the impact of outliers. We randomly select one-third of the railway and waterway locations, and assume that no hubs are available at those locations. Hybrid hubs are unavailable at the beginning of the model. The annual budgets are randomly selected using a uniform distribution $U(\$5, \$15)$ billion. The capacities are randomly selected using a uniform distribution $U(2, 10)$ million tons. For the 256-node network, only one instance is generated according to estimated data. This instance is also used to calculate the long-term large-scale projections in the later sections. Specifically, in the 256-node network, all existing facilities are available at the beginning of the model, whereas hybrid hubs are unavailable. The budget is set to be \$20 billion in the first year, and increases by 3% accounting for the inflation in the following years. The hub capacity is set to be 100 million tons per year for each facility. The runtime limit for the 256-node instance is set to be 12 h.

Objective, explored nodes, MIP gap, number of cuts, and runtime are collected from solving the instances using Gurobi. MIP gap refers to the optimality gap computed by Gurobi, where a gap of 0.00% indicates an optimal solution. The objective gap (Obj. gap) for B&C is the gap between B&C objective and MIP objective. Improvements of B&C runtime are computed based on MIP runtime. For 26, 39, and 80-node instances, the default MIP gap of 1×10^{-5} is used. For the 256-node instance, the MIP gaps for both MIP and B&C are set to be 1×10^{-2} to encourage algorithm convergence. This will not significantly impact the optimality, as the objective is on the scale of billion dollars.

Results are shown in Table 3. For most instances, MIP is able to reach the optimal solutions with 0.00% optimality gap, except for a

few instances where MIP cannot terminate within 2 h, with relatively small gaps ($< 0.01\%$). B&C solves all instances with 0.00% gap from MIP solutions. This validates the correctness of the B&C algorithm. For small instances, i.e., the 26-node dataset, B&C reaches the optimal solution 8.96% faster than MIP. As instances grow larger, B&C shows increasing improvements at 17.89% for the 39-node dataset and 84.14% for the 80-node dataset, respectively. For the 256-node network, MIP fully utilizes the 12-hour limit without producing a feasible solution. B&C identifies a suboptimal solution with 0.74% MIP gap within 7 h, representing a 41.20% improvement from the 12-hour time limit. Further, in Table 3, some instances take longer than others to solve, e.g., instances 3 and 4 of the 80-node network. This is primarily due to the variations when generating the instances, where the randomness in initially opened facilities, annual budgets, and hub capacities plays an important role in determining the hardness of the computation. Nonetheless, the overall results between 26, 39, and 80-node instances are consistent. Some of these factors will be further investigated through sensitivity analyses in the following section.

In addition, the numbers of explored nodes and added cuts are also reported in Table 3. These results demonstrate the computational advantage of the B&C approach, especially as the problem size grows. Essentially, B&C adopts a divide-and-conquer strategy, where a large-scale MIP is decomposed into easier-to-solve SPs. This can be observed from the number of explored nodes. Although B&C explores much more nodes than MIP, each node is solved faster due to the smaller-scale decomposed linear programs, resulting in overall more efficient solution procedures than MIP. The results further highlight the contribution of the B&C algorithm over classical Benders decomposition. In classical Benders, the cuts are added to the integer MP, instead of each nodal problem, which requires the MP to be solved repeatedly using the default B&B algorithm. As more and more cuts are added, solving each MP requires increasingly more nodes to be visited, eventually prolonging algorithm termination and wasting computational resources. Note that for the large 256-node network, the explored nodes are less than those of 80-node instances, due to the larger MIP gap, i.e., the solver may spend more effort closing the final 0.01% gap than finding a near-optimal solution.

5.2. Sensitivity analyses

Among model parameters, hub capacity and available budget significantly influence long-term model performance, but are difficult to estimate accurately due to a lack of historical data. As highlighted by Javaid et al. [84], limitations in capacity, particularly in cost-effective and sustainable modes such as rail and waterways, can act as a critical limitation to achieving efficient and low-emission freight flows. Likewise, the allocation of financial resources plays a crucial role in shaping investment priorities. Without proper budgetary planning, the intermodal system could be subject to inefficient operations and

Table 3
Algorithm comparison between MIP and benders decomposition with B&C.

Network nodes	Instance	MIP				B&C					
		Objective (\$Billion)	MIP Gap (%)	Explored Nodes	Runtime (s)	Obj. Gap (%)	MIP Gap (%)	Explored Nodes	Added Cuts	Runtime (s)	Impr. (%)
26	1	292.25	0.00	1	6.70	0.00	0.00	9	282	5.93	11.55
	2	292.78	0.00	1	6.43	0.00	0.00	1	222	5.70	11.40
	3	292.11	0.00	1	6.38	0.00	0.00	15	193	5.86	8.24
	4	292.37	0.00	1	6.92	0.00	0.00	7	278	6.04	12.66
	5	292.35	0.00	1	6.01	0.00	0.00	1	287	5.96	0.93
	Ave.	292.37	0.00	1	6.49	0.00	0.00	6.60	252.40	5.90	8.96
39	1	643.16	0.00	164	41.14	0.00	0.00	3633	1024	31.02	24.59
	2	662.62	0.00	991	50.29	0.00	0.00	8150	1097	41.93	16.62
	3	641.11	0.00	510	44.11	0.00	0.00	2090	1083	39.60	10.21
	4	655.80	0.00	218	40.04	0.00	0.00	2304	465	27.36	31.66
	5	685.93	0.00	1	31.60	0.00	0.00	965	607	29.59	6.37
	Ave.	657.73	0.00	376.80	41.43	0.00	0.00	3428.40	855.20	33.90	17.89
80	1	1964.57	0.00	4194	4044.62	0.00	0.00	137 871	2113	542.24	86.59
	2	1967.12	0.00	3137	5677.74	0.00	0.00	136 907	1639	508.35	91.05
	3	1978.88	<0.01	4481	7351.47	0.00	0.00	295 850	2976	1394.99	81.02
	4	1967.76	<0.01	4699	7352.12	0.00	0.00	551 087	3754	1762.48	76.03
	5	1963.94	0.00	6065	6724.32	0.00	0.00	284 878	3284	939.67	86.03
	Ave.	1968.45	0.00	6230.05	4515.20	0.00	0.00	281318.60	2753.20	1029.54	84.14
256	1	–	–	1	43200.00	–*	0.74	53 605	1184	25403.30	41.20

*: The obtained objective for the 256-node instance is \$11757.91 Billion.

high carbon emissions. In addition, carbon taxes have yet to be widely adopted in the U.S., where existing measures heavily depend on auction mechanisms [81]. We will further investigate model behavior under different carbon taxes.

First, we identify a base case for the sensitivity studies. The base annual budget to construct new intermodal facilities is set to be \$20 billion. This is consistent with the Infrastructure Investment and Jobs Act (IIJA) that provides approximately \$200 billion for new road, rail, and water transportation projects over a 10-year period [57]. The base annual hub capacity is set to be 20 million tons. This is estimated as the median of the annual tonnage of the top 50 U.S. ports [63]. As discussed in Section 4, the base carbon tax is set to be \$0.00005 per gram [81]. Note that due to the lack of data at commercial facilities, it is difficult to consistently estimate the freight capacity of rail stations. Also, historical annual tonnages may not accurately reflect the maximum capacity in the future. As such, the following sensitivity analyses provide insights for researchers and practitioners on the impact of these parameters on model behaviors.

Budget. To evaluate the sensitivity of the model to budget constraints, the yearly budget B_t , $t \in T$, is drawn from the set $\{\$5, \$10, \$20, \$30, \$50\}$ billion. The experiment is conducted using the 80-node dataset, and the problem is solved using B&C. For each budget scenario, we record the average annual transportation cost and the percentage ton-miles of road transportation. All other model parameters are fixed.

The results are shown in Fig. 3. The results show an inverse relationship between the budget level and the annual cost. As the budget increases, the model is able to construct and operate more hubs. This improves the network's flexibility, allowing more cost-effective shipping methods to be selected, eventually reducing the transportation cost. In addition, thanks to a strengthened intermodal infrastructure, the reliance on trucking is alleviated to some extent. Fig. 3 shows a continuous reduction in the percentage of trucking ton-miles, but the overall percentage change, around 0.5%, is not significant.

Fig. 3 also suggests that increasing the budget beyond a threshold does not lead to additional significant improvements, e.g., around 20 billion in our instance. This finding has practical implications for investment planning. It suggests that allocating a budget beyond the identified threshold does not yield significant operational savings, and decision-makers can prioritize reaching this optimal range without utilizing extra financial resources. The “plateau” in the costs indicates

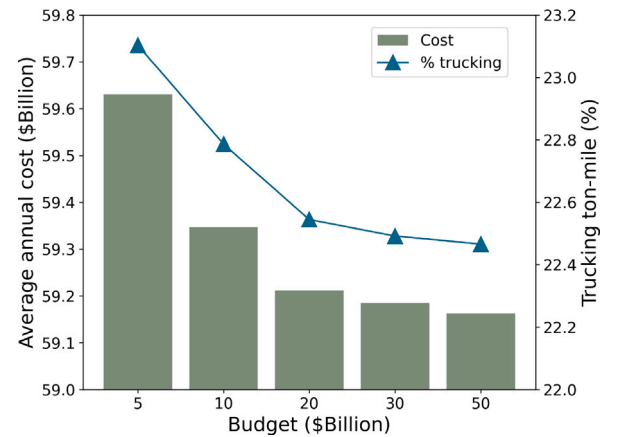


Fig. 3. Average annual cost and percentage of road transportation under varying budgets.

that most of the cost-effective hubs are already constructed and operated by that point, and additional hubs contribute marginally to network efficiency. Thus, further increases in budget would mainly enhance network redundancy without significantly improving total cost performance.

Capacity. Next, we vary the annual capacity of rail, water, and hybrid hub in the set $\{10, 20, 50, 100, 200\}$ million tons. All other model parameters are fixed. The model is solved using B&C.

The results are shown in Fig. 4. Similar to the budget, increasing hub capacities significantly reduces the transportation cost. Note that such a reduction has already incorporated the increased volume in each hub, as hub operating costs are calculated by the annually processed tonnage. As capacity increases, hubs handle more cargo, which also drastically reduces the reliance on trucking. Fig. 4 shows a near 40% reduction by planning for hubs with large capacities. There also exists a threshold, beyond which the improvement in transportation costs becomes minimal, e.g., from 100 to 200 million tons in Fig. 4. Also note that an annual capacity of more than 200 million tons may not be realistic for most facilities, as the busiest international ports (e.g. port

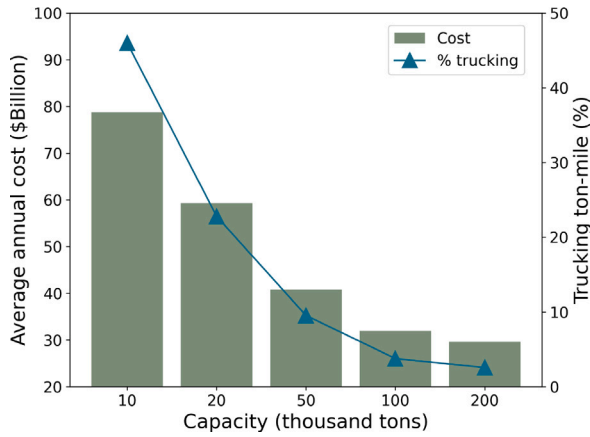


Fig. 4. Average annual cost and percentage of road transportation under varying capacities.

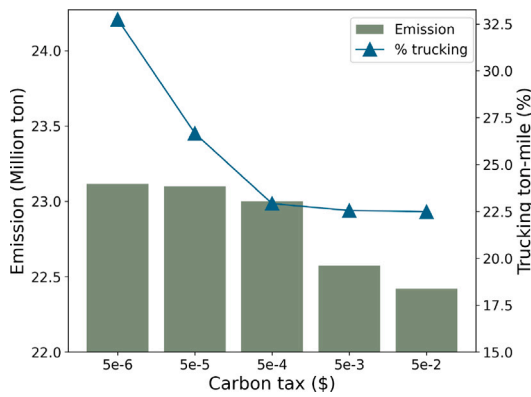


Fig. 5. The average annual emission and percentage of road transportation under varying carbon taxes.

Houston, port of South Louisiana) in the U.S. handles around 200–250 million tons [63].

These observations emphasize the importance of proper sizing of hub capacities in strategic infrastructure planning. Allocating sufficient capacities to meet long-term demand is crucial for achieving cost-effectiveness in network designs. At the same time, excessive investments in overcapacity yield diminishing returns both in operational performance and in computational effort. Identifying this capacity threshold early in the planning process can efficiently guide infrastructure investments and ensure robust network performance without unnecessary resource expenditure.

Carbon Tax. In addition, we investigate the impact of carbon taxes. We vary the carbon tax in the set $\{\$5 \times 10^{-6}, \$5 \times 10^{-5}, \$5 \times 10^{-4}, \$5 \times 10^{-3}, \$5 \times 10^{-2}\}$ per gram of emitted CO₂ equivalent. All other model parameters are fixed. The model is solved using B&C.

The results are shown in Fig. 5. Overall, the results suggest that increasing carbon taxes reduces annual emissions. Specifically, in the figure, increasing the taxes from $\$5 \times 10^{-6}$ to $\$5 \times 10^{-2}$ leads to around 600,000 tons of reduction in carbon emissions. In addition, increasing carbon taxes decreases the reliance on trucking by up to 10%, since rail and water transportation emit less GHG per ton-mile compared with trucking. As such, Fig. 5 provides insights for carbon tax policy-making. Too small a carbon tax rate, e.g., $\$5 \times 10^{-6}$, may not be effective in reducing emissions and promoting intermodality. Although a high tax rate, e.g. $\$5 \times 10^{-2}$, can significantly reduce emissions and the reliance on trucking, it may also impose a heavy burden on domestic businesses, lowering productivity. A moderate carbon tax rate, e.g., $\$5 \times 10^{-4}$ or

$\$5 \times 10^{-3}$, can effectively reduce emissions and provide the same level of incentives for infrastructure development, while avoiding negative impacts on economic prosperity in the transportation sector.

Elasticity. Here, we conduct experiments to investigate the elasticity of road, rail, and water transportation under varying costs. Specifically, we use the 80-node network instance and change the cost of road, rail, water transportation by -50%, -25%, 0%, 25%, and 50%. We collect the percentage of average annual total ton-miles of each mode, and calculate the cost elasticity using the percentage of change in ton-miles divided by the percentage of change in cost. A number close to 0 suggests inelastic mode usage toward transportation cost.

Results are shown in Fig. 6. Overall, results show that the annual ton-miles are inelastic toward costs. This is mainly due to the significant differences between road, rail, and water transportation costs. As estimated in Section 4, even with a 50% reduction, road transportation is still more expensive than rail and water. The inelastic result is consistent with empirical studies in the literature [85,86]. In particular, rail is slightly more elastic than road transportation, and water transportation is the least elastic among all three [85]. The inelastic behavior emphasizes the strength of intermodal transportation, where rail and water transportation significantly reduce costs over long distances for bulk cargo, whereas road transportation can flexibly reach destinations with minimal infrastructural support.

5.3. Future projections

In this section, we use projected freight volume and emission data to study the impact of intermodal transportation for the next two decades. The instance constructed from the full 256-node dataset, discussed in Section 5.1, is solved to provide the projections. First, we consider the scenario where only existing infrastructure is available to investigate transportation mode shifts. Second, we evaluate the benefit of infrastructure investment in hybrid hubs. Finally, we look at the impact of clean vehicle technologies.

Existing Infrastructure. Here, we assume that no new intermodal infrastructure is developed. We construct an intermodal network using the full 256-node dataset, where hybrid hubs are disabled when analyzing the existing Infrastructure. Freight volumes in FAF from origin to destination during the years 2025, 2030, 2035, 2040, 2045, and 2050 are included. A benchmark is further constructed from FAF, where the percentage of rail and water transportation is set to be in line with FAF projections, i.e., 11%–12% of freight uses rail and 8%–9% of freight uses water. An annual inflation of 1.03 is also accounted for in the cost.

In contrast, in our model, the usage of intermodal transportation is fully optimized, under the constraint of existing infrastructure, e.g., hub capacity, availability of connections, routing distance, etc. The comparison between our model and the FAF benchmark is shown in Fig. 7. Results indicate a significant reduction in both transportation cost and carbon emissions. Particularly, the increased intermodality from 2025 to 2050 saves 51.55–137.09 billion in transportation costs, as well as 5.57–15.48 million tons of CO₂ equivalent GHG emissions. Note that although the optimized results may not be immediately applicable due to the gap between computational studies and real-world operations, the results still provide valuable insights into what could be achieved if the transportation industry reduces its reliance on trucking.

Hybrid Intermodal Hubs. To demonstrate the practical application of hybrid intermodal hubs and validate our model's ability to support strategic decision-making, we solve a large-scale infrastructure development problem over the next two decades. The 265-node dataset is used. We assume that the accompanying railway connection costs are included in the hub construction costs. Interpolated annual demand data from FAF are utilized to calculate infrastructure development and freight transportation from 2025 to 2050.

Specifically, we control the variables in this experiment to ensure that the differences in cost and emission metrics come solely from the additional hybrid hubs. As such, this experiment evaluates the

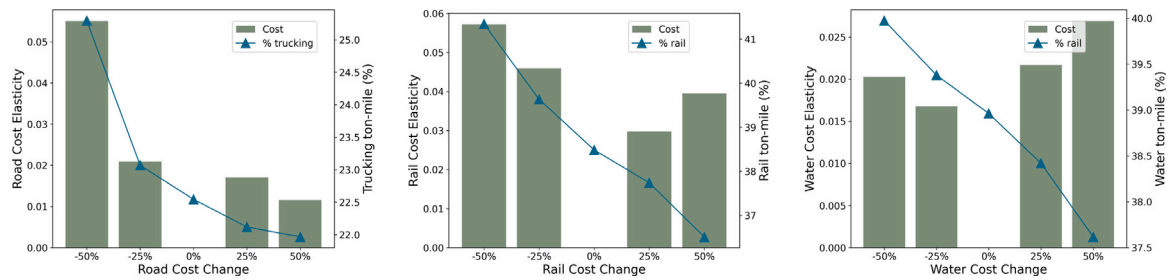


Fig. 6. Average annual ton-mile elasticity under $\pm 25\%$ and $\pm 50\%$ changes of road, rail, and water transportation costs.

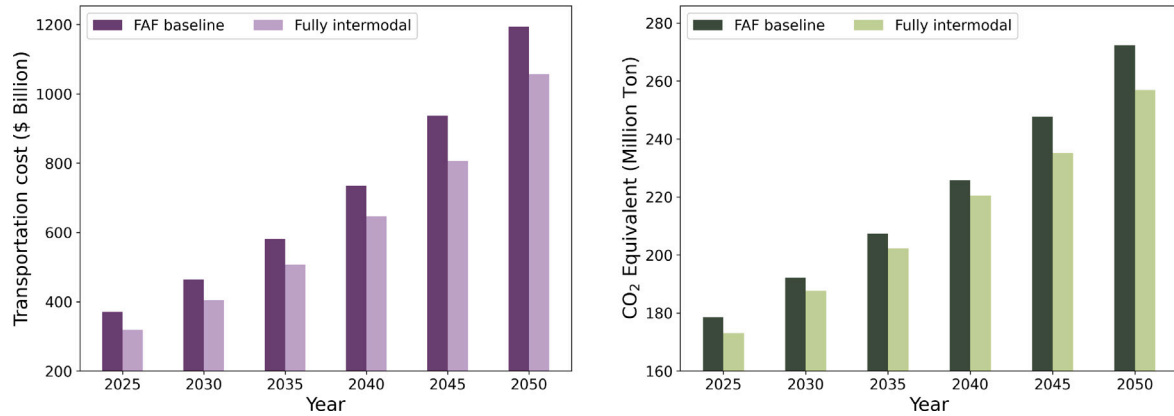


Fig. 7. Comparison of cost and emissions between FAF projections and fully optimized intermodal transportation.

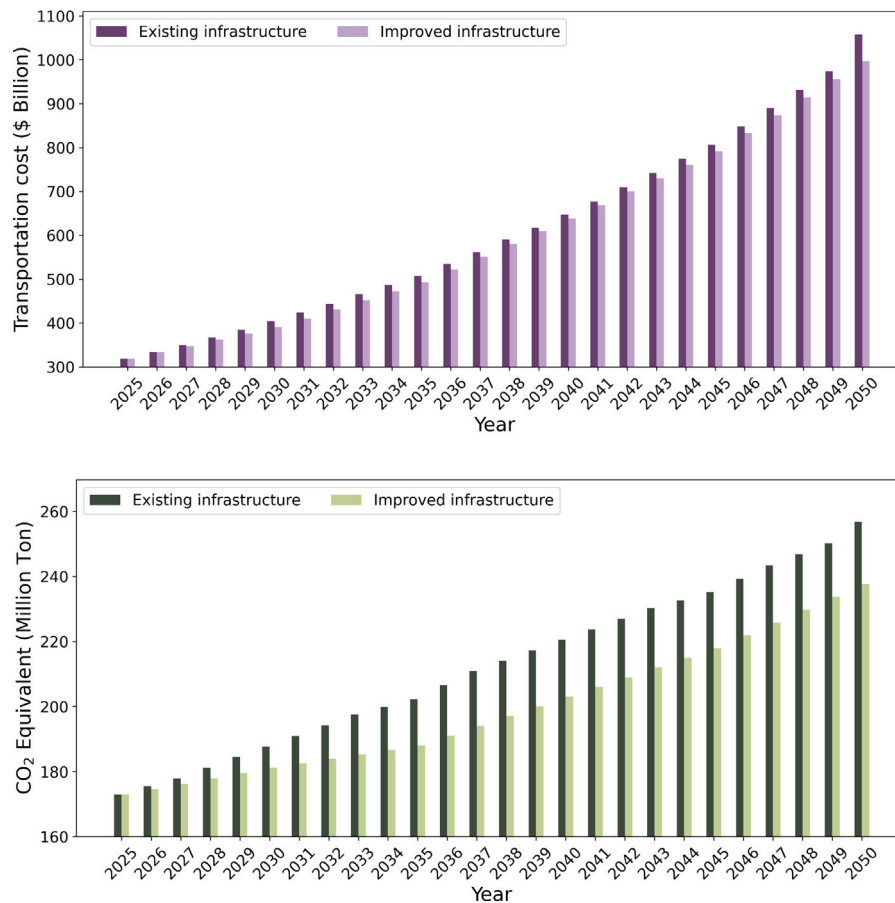


Fig. 8. Comparison of cost and emission between existing and improved infrastructure.

Table 4
Intermodal hub construction, investment, and return over a 26-year horizon.

Year	Hybrid hubs constructed	Hub cost (\$Billion)	Transportation cost saved (\$Billion)	Return on investment (%)	GHG emission saved (Million tons)
2025–2030	28	72.00	28.80	–60.00	16.97
2031–2035	9	27.64	98.04	27.29	58.44
2036–2040	1	6.29	99.31	113.48	83.93
2041–2045	2	11.28	107.86	184.94	88.89
2046–2050	0	0	146.97	310.32	87.61
Total	39	117.21	480.98	310.32	335.84

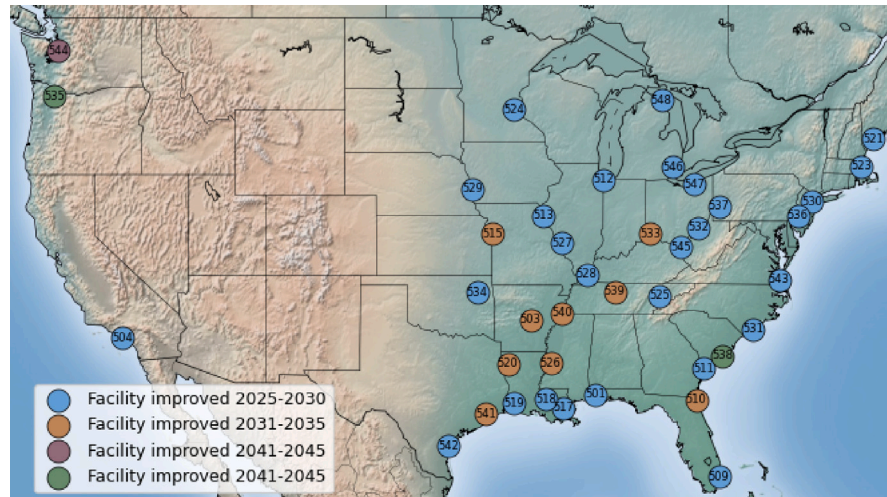


Fig. 9. Hybrid hubs constructed during 2025 to 2050.

long-term nationwide impact of hybrid hubs, as well as their potential for intermodal infrastructure investment projects. The fully optimized intermodal baseline from the previous study is used as the benchmark and is re-solved during the 26-year horizon to obtain annual results, which feature existing infrastructure without hybrid hub capacities. Transportation costs and carbon emissions are collected and compared in Fig. 8. Results show that through intermodal infrastructure development, transportation costs can be further decreased by \$1.26–\$60.20 billion annually. This is also accompanied by 1.57–19.13 million tons of reductions in CO₂ equivalent GHG emissions.

In addition, a summary of collected metrics is presented in Table 4. Overall, our model recommends heavy infrastructure investment at the beginning of the planning horizon, due to their long-term cost-saving benefit and cheaper construction costs not subject to inflation. The return on investment calculates cumulative returns from saved transportation costs over the hub construction and fixed operating costs. Due to initial heavy investment, the return on investment is negative in the first five years, but gradually turns positive as the infrastructure accumulates more savings in transportation costs. Over the 25-year horizon, the total return on investment reaches around 310.32%. The return on investment is further validated by several sources in the literature on transportation infrastructure development, which suggest up to 100%–400% return on investment [87,88]. Our result is roughly in line with the existing estimations. The constructed hybrid hubs are further illustrated in Fig. 9. The results provide guidelines and insights for practitioners regarding the value of intermodal infrastructure development. Particularly, facilitating transshipment between railway and waterway via hybrid hubs leads to substantial economic benefits. Although the majority of transportation cost-saving benefits the private sectors, government institutions will still have significant returns from their investment. Higher profit margins in individual freight companies lead to more corporate income taxes to the federal and state government. A booming transportation sector contributes to the overall economic property in commerce, manufacturing, supply

chain, finances, etc. Furthermore, carbon emission taxes create new avenues for profit sharing with the public sector.

Impact of Clean Vehicle Technology. In this section, we evaluate the impact of clean vehicle technologies on reducing GHG emissions in the long term. To achieve this, we assume that the future investment in clean vehicle technologies, e.g., electric, hydrogen, fuel cells, etc., can achieve the recommendations from the International Energy Agency's (IEA) Net Zero by 2050 report, with the goal to realize carbon neutrality in 2050 [83]. The technological maturity from IEA is matched with vehicle and fuel-specific emission data in the GREET software to reflect the case context in the U.S. [78]. The IEA report provides data on fleet and energy compositions necessary to achieve carbon neutrality by 2050. Specifically, oil-based fuels are projected to be of approximately 75%, 44%, and 17% in transportation energy demand in 2030, 2040, and 2050, respectively. Based on such composition, the emission data of road, rail, and water transportation are estimated from the GREET software.

The results are shown in Fig. 10. The status quo is calculated from the solution of the improved intermodal infrastructure in the previous experiment. Despite increased freight volumes, carbon emissions can be significantly reduced when clean vehicle technologies are introduced to the transportation sector in the following decades. Specifically, from 2025 to 2050, an average of 73.56 million tons of CO₂ equivalent GHG emissions can be saved annually, totaling 1912.45 million tons throughout the 26-year horizon. The result signifies the importance of early investment in clean vehicle technologies, as well as their market adoption in the transportation industry.

To summarize, the above experiments utilize real-world data to calculate projections of U.S. transportation trajectories in a 26-year horizon. Together, the experiments validate the effectiveness of our model in predicting system-wide metrics using a variety of data sources. The results show that the joint approaches of freight flow optimization, infrastructure development, and clean vehicle investment can transform the transportation industry by bringing substantial savings in

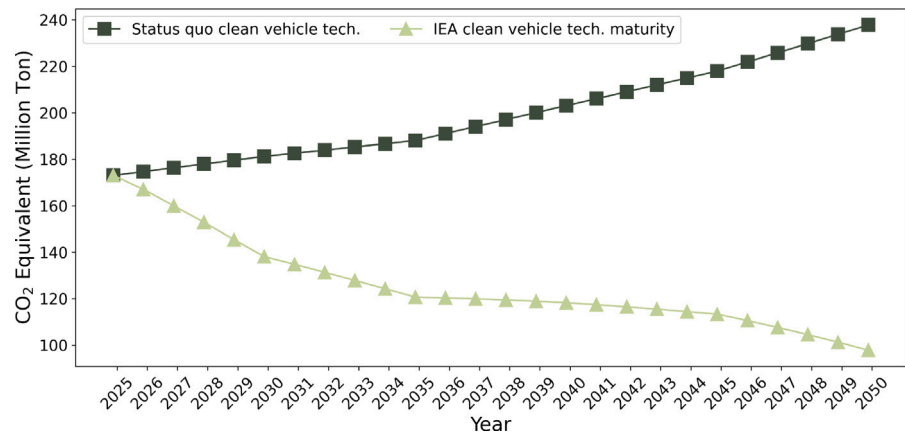


Fig. 10. Comparison of carbon emission between status quo in clean vehicle technology and the maturity based on IEA’s Net Zero by 2050 plan.

Table 5
Abbreviations and definitions in this study.

Abbreviation	Definition
B&B	Branch-and-Bound
B&C	Branch-and-Cut
EPA	Environmental Protection Agency
FAF	Freight Analysis Framework
FTOT	Freight and Fuel Transportation Optimization Tool
GHG	Greenhouse Gas
GREET	The Greenhouse Gases, Regulated Emissions, and Energy Use in Transportation Model
IEA	International Energy Agency
IIJA	Infrastructure Investment and Jobs Act
MIP	Mixed Integer Programming
MP	Master Problem
SP	Subproblems
TEU	Twenty-foot Equivalent Units

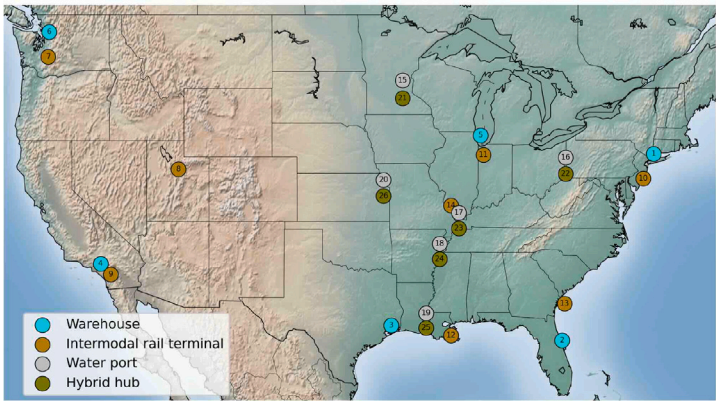


Fig. 11. Facilities in the 26-node sub-network.

operating costs and reductions in carbon emissions. Practitioners will be able to draw insights from existing results or generate new ones to inform their decision-making processes.

6. Conclusion

This research develops a strategic optimization framework for improving intermodal freight networks through integrating hybrid hubs. The results demonstrate the significance of our proposed approach,

reducing total operating costs and GHG emissions compared with traditional truck-only or single-mode transportation networks. The model supports sustainable and cost-effective freight movement by allowing freight to seamlessly transfer between road, rail, and water transportation. Our experiments show that hub capacity and budget allocation are key factors affecting network efficiency and the reliance on trucking. Strategic planning, including the timing and siting of hybrid intermodal hubs, significantly enhances network efficiency and flexibility, leading to cost and emission reduction. Specifically, well-optimized

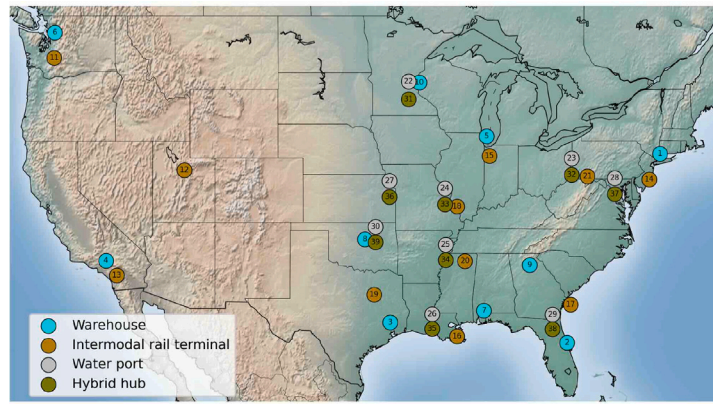


Fig. 12. Facilities in the 39-node sub-network.

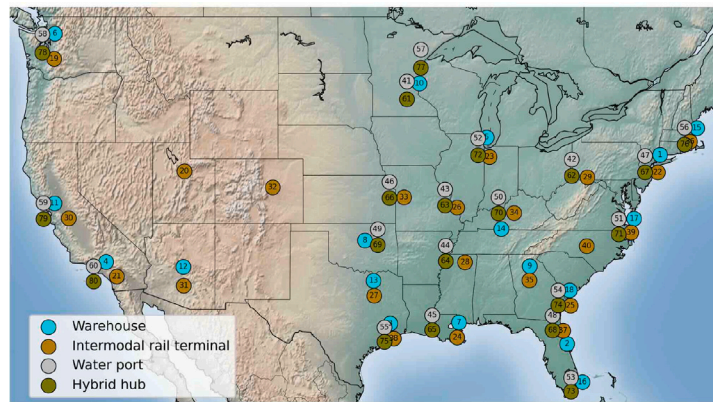


Fig. 13. Facilities in the 80-node sub-network.

transportation infrastructure investment could realize over 200% return on investment during a 26-year horizon. Fully capitalizing on the maturing clean vehicle technologies could reduce carbon emissions by 54 million tons at an annual rate. In addition, our B&C decomposition algorithm effectively reduces computational complexity and runtime, by up to around 50% on medium instances, which enables us to conduct further numerical studies at the national scale. Overall, our study provides actionable insights into national-level intermodal freight transportation. Our study also supports the development of an environmentally-sustainable and operationally-efficient intermodal infrastructure network, demonstrating the significance of long-term planning.

There are several ways that this research can be extended and improved. For example, a larger profile of strategic decisions can be incorporated into the model, such as capacity expansion at existing facilities, the construction of new linkages, and upgrading existing linkages to improve capacity. Tactical decisions, such as fleet composition, vehicle routing, freight transshipment, and hub inventory management, can also be integrated into the model. These decisions may render a more complex model that requires additional algorithmic innovation. Furthermore, future work can explore uncertainties in yearly freight volumes, facility operating efficiency facing disruption, or the availability of transportation routes due to disasters. These scenarios demand pre-disruption planning and dynamic recourse actions afterwards. New modeling techniques and algorithm development will be necessary to produce good-quality solutions. Last, additional practical factors can be incorporated to strengthen the real-world case studies. Existing infrastructure development projects can be included to further investigate the impact of intermodal infrastructure development. Effort is need to overcome the challenge in the availability of data and the uncertainty

of construction delay or even cancellation. As clean vehicle technologies begin to commercialize, real-world data can be collected to enable endogenous modeling of fuel choices and availability, instead of using averaged parameters, which will further enhance the credibility and practicality of nationwide long-term projection results.

CRediT authorship contribution statement

Ayoub Abusalih: Software, Methodology, Data curation. **Zeyu Liu:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zeyu Liu reports financial support was provided by Advanced Research Projects Agency-Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Abbreviations

See Table 5 for a summary of abbreviations and definitions used in this study.

Appendix B. Selected network nodes

Figs. 11, 12, and 13 shows the selected intermodal transportation facilities for the 26, 39, and 80-node networks. Slight offsets are applied to the nodes to avoid complete overlaps.

Data availability

The data is available in an online repository.

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